7-1 Final Project

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The SNHU Travel application was to have a team use a Scrum-Agile approach in developing a software that allows the travel agency to extend its client base by using innovative tools. As a company considers transitioning from a traditional waterfall development model to an Agile methodology using Scrum framework, trying it out on a smaller scale first allows for a company to see its true benefits when done right. As the Scrum Master, I am making this Sprint Review and Retrospective to give a summary on the outcome of the work that has been done.

My role as the Scrum Master, meant I was in charge of facilitating Scrum meetings and starting them on time and preventing them from dragging on longer than the initial time frame. I was also responsible for creating the Agile team charter which provided information regarding aspects of the SNHU Travel application. For instance, a description of what the software was and what the task was intended to perform for the end user, dates on when the project would start and when it would be completed, a list of some of the key project objectives, mentioning of risks that could hinder the project, and most importantly rules of behavior and communication guidelines were set in place for the team to stay aligned with one another. I was also responsible for fixing issues such as removing impediments which would have prevented the team from doing their job, jeopardizing the flow of the project.

The Product Owner was in charge of keeping up good communications with the stakeholders/customers in order to ensure that the project was still aligned with the wants of the customer and then relaying any new information back to the development time, that is if the new items are reasonable. Another job of the product owner was to prevent any impossible tasks from being added into the project which would hurt the development team, so they protect the development team but ensuring that only tasks of high value and that can be accomplished within the time frame for completing the project are implemented. The Product Owner was also

responsible for creating User Stories for the development team which provides an easy to understand and clear way of describing each feature and what the feature is intended to do. This gives the development team the reason why a feature is being made, but without telling them how to create it.

The Testers were not just there to test the code for bugs and prevent the software from being released with issues, making the application not very user friendly, but they were also responsible for creating test cases. These test cases were created with the User Stories that the Product Owner made, and their main purpose is to show what you can do for a feature and the result of that feature. This would showcase what the feature should perform, making the testing of the code easy as you have listed out exactly what it should do. The Testers would communicate through email with questions for the Product Owner regarding specific details of the features to create the test cases, that way they could better understand what needed to be achieved by the feature.

The Developers were in charge of creating the code for the software, ensuring it was clean and easily scalable if need be. Throughout the Sprints, the Product Owner met with the stakeholders and got new information regarding the content within the SNHU Travel application, and so the change was relayed onto the developer team. They make the necessary changes for the application to fit more with what the customers will be looking for. For example, the application was focused on any travel however it changed to detox and wellness travel. Developers quickly changed colors, styles, and implemented a new list viewing system of the destinations to fit with the slide show look the project now required.

A Scrum-Agile approach was very beneficial in the creation of the user stories. By having a meeting with the stakeholders/customers we were able to figure out exactly what they wished

to see in the application, helping us to understand what their needs are for the software. Getting this information helped the Product Owner make user stories that were tailored to the customers needs and formatted in a way anyone could clearly understand. By also interacting with the stakeholders/customers, we were able to determine the priority of each item and the overall size whether it was small or large in scale. Each user story value statement was written clearly, using the format of as a(n) <type of user>, I want to <perform some task>, so that I can <achieve some goal>, and then all acceptance criteria was written for each story. The development team was then able to easily use the stories to start building the application and the code that was needed for the software to function in the way that the customers would want it to. Instead of just guessing what the customer wants and needs, this Scrum-Agile approach ensures they understand the goal before writing any lines of code. For instance, the top five destinations user story had a list of criteria that needed to be there in order to be complete. In a waterfall approach, you would wait till the very end to test the whole application, but in this Agile approach, it focuses you to think about how to prove the story is done and know for a fact five destinations are displayed, each with pictures of their own, descriptions about the destination and activities to do there, and a link to send the customer to a new tab to view booking options. Changes were able to be made based on feedback received from the stakeholders/customers, specifically for displaying the top five destinations and how at first it was meant to just have the user scroll through a list of the destinations, but was later changed to a slide show.

When changes or interruptions happen within the project, the team was able to be flexible around the changes as the work is done in short cycles and the entire project plan does not have to be reviewed and only what is affected by the change. Like when the change to the design for displaying the top five destinations was swapped from a scroll design to a slide show, the Scrum-

Agile approach allows us to take the new requirement and turn it into a new user story and prioritize it within the next Sprint. If this was a waterfall approach, a change like this would not be done till the very end of the project, which might require a renegotiation with the stakeholders, pushing the release of the application by months because of replanning. Another example would be when the stakeholders/customers wanted to focus on a detox and wellness travel plan for the top five destinations instead of just overall travel locations. This was an interruption for the project however the team handled it by updating user stories and prioritizing the new detox and wellness theme and modifying the destinations by performing backlog refinement. The waterfall approach would have been a disaster because the months of work you did now has to be redesigned and you must start from scratch, but with the Scrum-Agile approach, you are able to make changes like this throughout the project without any major issues to the flow of the project.

Throughout the project the team communicated through daily Scrum meetings to discuss what they have worked on, what they plan to do next, and if they had any impediments. This allowed the team to see overall where the project stood and what has been done so far. The developer communicated to the product owner and tester via email to ask questions regarding the functionality and layout of certain features of the program to ensure they were creating code that fit with the requirements before starting on something and wasting time. A discussion board was used to discuss which Agile practices we should focus on and a place where we could ask questions on which practices were most beneficial to the project.

The product backlog and sprint backlog was a very useful tool in prioritizing the detox and wellness requirements over the general travel we originally had, as it kept the goal of the SNHU Travel application visible and did not let us get distracted by other features that were not

yet needed. User stories and test cases helped the developers create the code efficiently while making it clean and scalable. The use of Eclipse for Java was also very beneficial and effective to use as our IDE to create our code in. The principles of transparency, inspection, and adaptation were helpful to the team's success by having within the user stories clear acceptance criteria. This helps to know when a feature is actually done. Showing the stakeholders in a sprint review is what we would do to showcase what was completed so far, instead of what was still in the backlog.

Pros of the Scrum-Agile approach would include how when the stakeholders/customers changed the travel to detox and wellness theme, and how Agile allows us to change easy and pivot to the following sprint. Agile is very flexible and changes can be made throughout the project without disrupting the flow. Unlike waterfall which would have wasted weeks of building the travel that would no longer be desired by the customers. The SNHU Travel application saw working code early on in the project, which ensures that trust is built between the team and the stakeholders for a successful outcome of the project. By writing test cases early in the development for the application, it allowed for the team to find logic errors in the code which prevents the risk of massive bugs that could happen later on in the project.

Cons of the Scrum-Agile approach would include scope creep. While Agile is flexible it could cause issues since you could keep adding features to the application causing there to never be an end date of the project. Also, stakeholders and customers may ask for features very last minute which makes them believe that the development team can still implement a huge feature in the last sprint, which would put stress on the development team if the feature was something important that was left out till last second. Another issue that could happen with Agle is the constant meetings like the daily meetings and sprint reviews and planning which could cause the

team to feel as though their time is being taken away from the actual coding for the application. To prevent something like this, I as the Scrum Master would have to ensure that meetings do not run longer than intended and that and impediments are discussed with the people who are directly affected by the impediments leaving the others who are not affected to keep work on the project.

Using the Scrum-Agile approach was the best choice for the SNHU Travel application development project as the market of travel is constantly changing and the customers will most likely be looking at what is trendy at that time. You would be able to focus on the market shifts to better tailor to customer needs and wants, like moving from the overall travel destinations to focus on detox and wellness instead. Waterfall if best suited for projects that will not have changes within them, so Agile is the best approach for this particular project. Since the applications for SNHU Travel will constantly be changing to meet with trends, Agile will ensure that the final product actually meets with customer wants when released to the public.